Liquidity Shocks in Virtual Markets

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Introduction

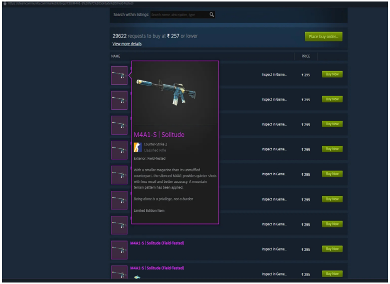
- This paper serves as a case study to estimate the effects of sudden market structure changes in virtual asset markets.
- In particular, I look at the market for cosmetics (AKA "skins") in the video games Counter-Strike and Dota 2.
    - The market for these items is surprisingly large: Counter-Strike hit a market capitalization of $4.3 billion in March 2025 (D'Anastasio 2025).
    - On 3/29/18, all Counter-Strike items were trade restricted, preventing sale & trade within 7 days of acquisition. This restriction continues today and was not applied to Dota 2 items until May 2020.
- I use a DID specification to identify a 4.2% decrease in quality-adjusted prices and 34.4% drop in trade volume after implementation of a 7-day trade ban on a subset of items listed on a virtual exchange.
    - Overall, the treated items maintained a higher price level than the untreated items, suggesting that these items retained their amenity value but lost investment value causing a dip in quality-adjusted prices.
    - This effect manifests as a downward level shift that remains significant for over a year post-treatment despite the growing number of users, and is consistent with empirical findings in the literature.

Contribution

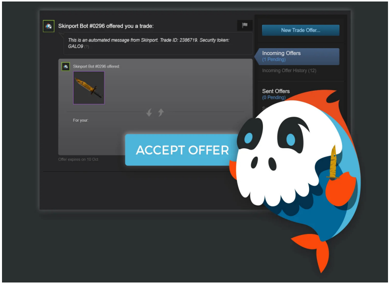
- **Treasuries & OTC Markets:** Trade delays and market driven declines in liquidity have meaningful effects on asset price and choice (Duffie, Gârleanu, and Pedersen 2005; 2007; Krishnamurthy 2002; Lagos and Rocheteau 2007; Amihud, Mendelson, Pedersen, et al. 2006; Geromichalos and Herrenbrueck 2016).
  - **Contribution:** I study a fixed top-down implementation of an exogenous trade delay, allowing me directly observe the liquidity premium.
- **Hedonic Pricing:** Unquantifiable properties intrinsic to goods (aesthetics, location, etc.) play an important role in pricing and consumer choice (Lancaster 1966; Rosen 1974; 1986; Sirmans, Macpherson, and Zietz 2005).
  - **Contribution:** I show this applies to virtual assets designed and governed by a singular entity. In this case, the price is largely driven by these aesthetic features.
- **Alternative Digital Assets:** As assets, Counter-Strike skins experience sizable returns, with average annual returns of 40% and low exposure to traditional financial markets. Lower financial returns on expensive skins are likely compensated by the "emotional dividend" of gamers (Dobrynskaya and Strelnikov 2025).
  - **Contribution:** I show that the emotional dividend is large as most of the skins' value is retained. I show that market structure is incredibly relevant for the growing number of virtual exchanges (eg: P2P Crypto Exchanges).

Skins Overview

The Marketplaces

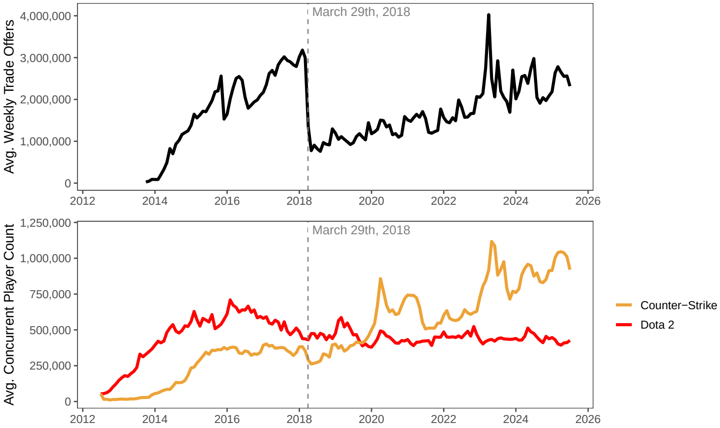


Steam Community Market Listings



Trade with a Third-Party Bot Account

The 3/29/18 Trade Ban



Note: Steam trade offer data includes trade offers for all Steam items and is estimated from the difference in trade offer ID between trades. Top chart is sourced from Trade.TF and bottom figure is sourced from Steam Charts .

Dataset

Counter-Strike and Dota 2 Items Summary

	Counter-Strike	Dota 2	Combined
Observations	2940	2675	5615
Unique Items	60	56	116
Oldest Origin Date	2013-08-14	2014-05-08	2013-08-14
Newest Origin Date	2016-02-18	2016-10-05	2016-10-05
Median Sale Price	\$4.26	\$1.90	\$4.13
	(\$7.28)	(\$3.61)	(\$7.14)
Log Median Sale Price	0.517	-0.064	0.484
	(1.379)	(1.160)	(1.253)
Volume Sold	20524.25	1356.84	11392.85
	(37685.37)	(2201.17)	(28938.78)
Grade Rarity	0.154	0.110	0.152
	(0.109)	(0.065)	(0.107)

Note: Median Price and Grade Rarity are monthly averages weighted by volume sold with weighted standard deviation in parentheses below. Volume Sold is a monthly average with unweighted standard errors. Origin date and grade rarity are provided by CS2 Database and Backpack.TF respectively.

Regressions

First Stage Hedonic Regression (Quality-Adjustment):

$$\log(p_{it}) = \beta_0 + \beta_1 ItemAge_{it} + \beta_2 ItemAge_{it}^2 + \beta_3 GradeRarity_i + \beta_4 GradeRarity_i^2 + \varepsilon_{it} \quad (1)$$

- We carry out this first stage regression to control for heterogeneity in item quality.
- Individual items can become more or less valuable due to changing taste preferences and/or game updates that modify in-game performance or appearances.

TWFE Regression:

$$\log(\hat{p}_{it}) = \beta_0 + \beta_1 \log(ConcurrentPlayers)_{it} + \delta\, Treated_i \times Post_t + \underbrace{\alpha_i}_{\text{item FE}} + \underbrace{\gamma_t}_{\text{month FE}} + \varepsilon_{it} \quad (2)$$

Event Study Regression:

$$\log(\hat{p}_{it}) = \beta_0 + \beta_1 \log(ConcurrentPlayers)_{it} + \sum_{k \neq -1} \delta_k \cdot \underbrace{\mathbb{1}(k = t - e)_t}_{\text{k months before treatment period e dummies}} \cdot Treated_i + \alpha_i + \gamma_t + \varepsilon_{it} \quad (3)$$

Regression Results

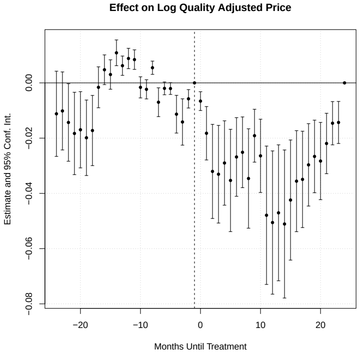
First Stage Hedonic and TWFE Regression Results

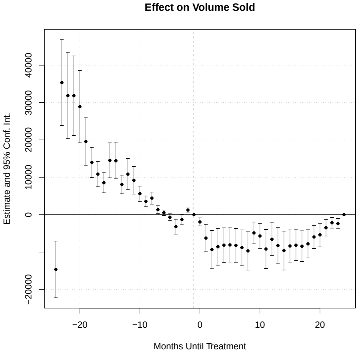
	$\log(p_{it})$
β_0	1.595^{***}
	(0.075)
ItemAge	0.029^{***}
	(0.003)
$ItemAge^2$	-0.001^{***}
	(0.000)
GradeRarity	-19.43^{***}
	(0.894)
$GradeRarity^2$	38.37^{***}
	(2.623)
N	5615
Adjusted R^2	0.452

	$\log(\hat{p}_{it})$	$\log(Volume\ Sold)$
$\log(ConcurrentPlayers)$	-0.050^{***}	-0.422^{***}
	(0.012)	(0.120)
$Treated \times Post$	-0.043^{***}	-0.326^{***}
	(0.010)	(0.081)
N	5579	5615
Adjusted R^2	0.997	0.956
Within R^2	0.130	0.055
Item and Month FEs	✓	✓

Note: * $p < 0.5$; ** $p < 0.01$; *** $p < 0.001$. Standard errors are shown in parentheses below their respective estimates. Both regressions with price as a dependent variable are weighted by Volume Sold. Item Age is in terms of months.

Event Study Results





Note: The last period, $t = 24$ is excluded and shown here as 0 due to collinearity.

Next Steps

- Additional controls to handle "seasonality" in game updates.
- Additional troubleshooting on event study design.
- Web scrape listed buy orders and offers from the Internet Archive's Wayback Machine.
    - Gives us the bid-ask spread which gives us a direct measure of the decline in liquidity.
    - May provide more insight into the liquidity premium price channel.
    - Only challenge (besides coding) is endogeneity in data gaps.
- Eventually a structural model may help me produce more interpretable estimates.

Conclusion

- Liquidity plays a key role in determining prices, returns, and choice in physical and virtual assets.
    - Despite being primarily valued for their aesthetic, Counter-Strike skins faced a 4.2% illiquidity discount and 34.4% decline in trade volume after trade ban implementation.
- As online exchanges proliferate, market administrators need to carefully consider market structure.
- Virtual markets can serve as valuable sources of high frequency data relating to more traditional ones.

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